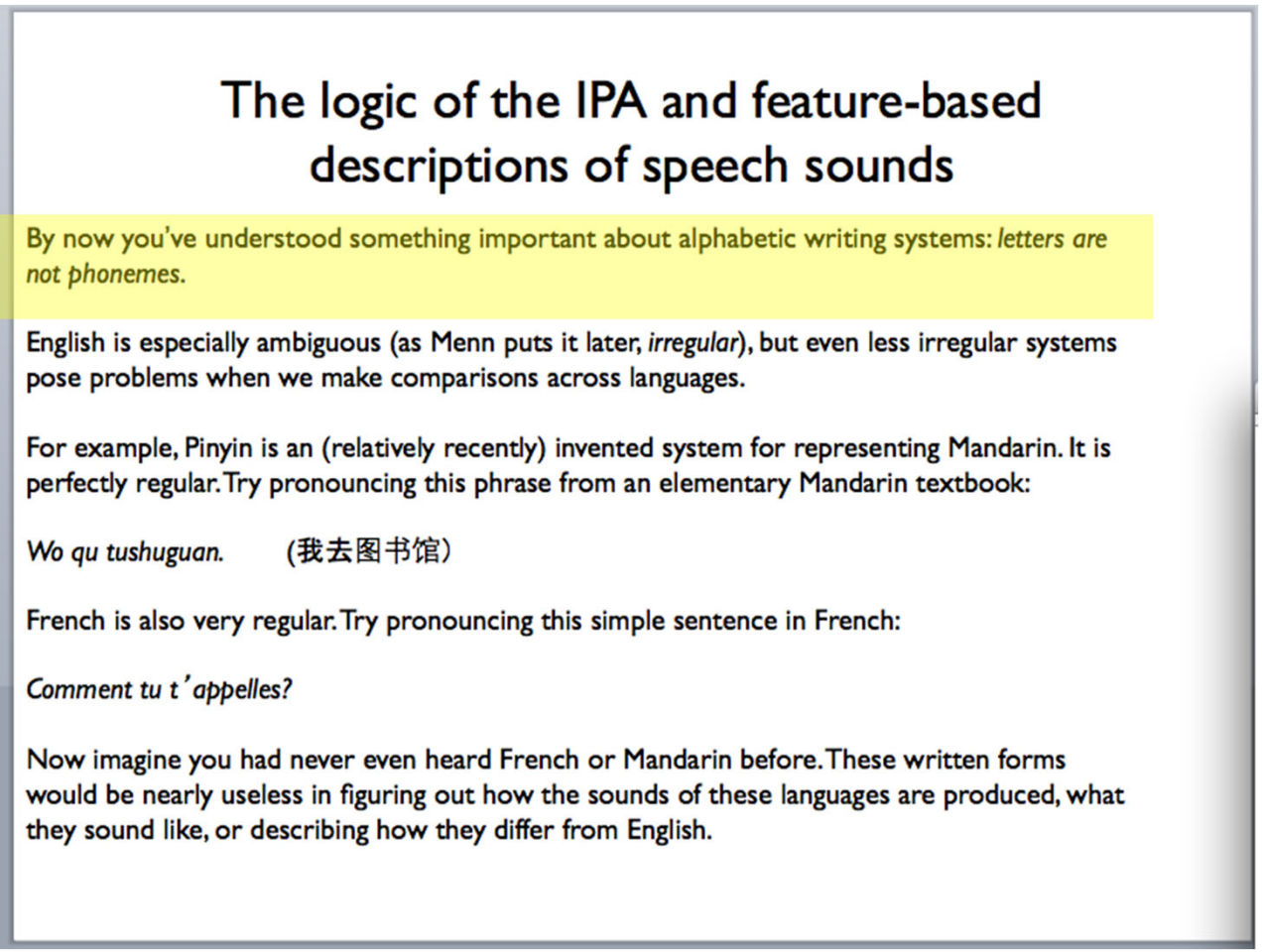
Reading 1: Empirical studies

Do you remember when we discussed phonemes at the beginning of the semester?

What was the first thing I wanted you to remember about phonemes?



The logic of the IPA and feature-based descriptions of speech sounds

By now you've understood something important about alphabetic writing systems: *letters are not phonemes.*

English is especially ambiguous (as Menn puts it later, *irregular*), but even less irregular systems pose problems when we make comparisons across languages.

For example, Pinyin is an (relatively recently) invented system for representing Mandarin. It is perfectly regular. Try pronouncing this phrase from an elementary Mandarin textbook:

Wo qu tushuguan. (我去图书馆)

French is also very regular. Try pronouncing this simple sentence in French:

Comment tu t'appelles?

Now imagine you had never even heard French or Mandarin before. These written forms would be nearly useless in figuring out how the sounds of these languages are produced, what they sound like, or describing how they differ from English.

We were able to demonstrate that phonemes are not letters, but also that *letters are not phonemes* with a "mispronouncing contest" in which people willfully mispronounced words like:

COLONEL

PARADIGM

We also talked about how some scripts (like Pinyin for Mandarin and French) are relatively *regular*.

This difference between scripts is also referred to as *orthographic depth*.

English, due its irregularities is a relatively *deep* script.

Finnish and Korean are very *shallow* scripts.

Listen and repeat:

00:00 00:00



Teija: Juan Ruiz?

Juan: Si! Kyllä, joo! Minä olen Juan Ruiz.

Teija: Hei, minä olen Teija, Teija Virtanen. Tervetuloa Suomeen!

Juan: Kiitos!

http://tasteoffinnish.fi/lesson1-1/

Some writing systems, like Finnish, are very “transparent” (or “shallow” according to the Orthographic Depth Hypothesis).

Let’s listen to this beginning Finnish lesson and follow along with the text…

After about a minute of this, you probably get the idea?

Each letter corresponds more or less unambiguously to one phoneme.

Doubled letters actually correspond to geminated consonants or long vowels.

Finnish children learn to *decode* pronunciations for written forms very very quickly.

Table 3. *Percentage of accurately read pseudowords (means and standard deviations)*

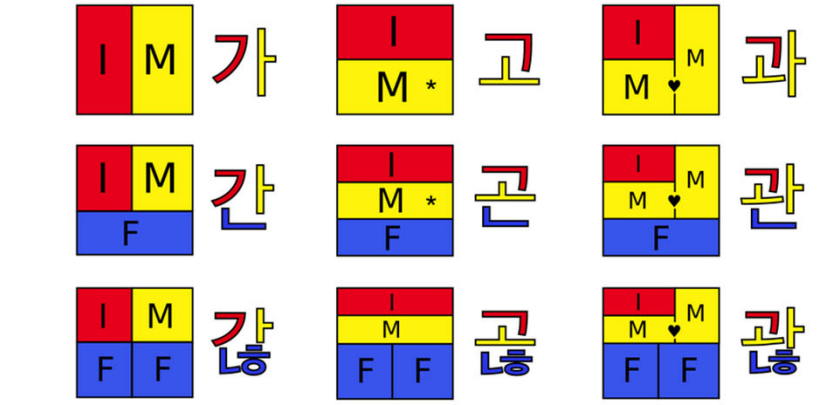
	Grade 1		Grade 2		Grade 3		Grade 4	
	M	*SD*	*M*	*SD*	*M*	*SD*	*M*	*SD*
English	50.3	32.8	71.0	32.5	73.5	28.8	88.2	15.2
French	86.7	5.9	96.7	3.5	98.4	2.4	98.5	2.7
German	88.0	12.4	87.3	9.5	86.0	15.2	87.2	19.5
Dutch	85.2	8.0	88.9	9.1	91.2	8.1	95.1	5.8
Spanish	87.3	12.2	90.4	8.1	90.9	6.8	90.6	8.2
Swedish	93.2	9.6	90.8	10.5	95.4	8.2	97.4	4.3
Finnish	84.9	13.6	89.6	10.1	88.4	8.1	93.7	8.3

In fact, English is a bit of an outlier among languages that use the Roman alphabet.

One result is that kids learn to decode more slowly in English.

In Grade 1, kids learning a wide variety of other languages are able to generate plausible pronunciations for "pseudowords," (made up words that follow the spelling and sound patterns of their language) at a much higher rate than English learners.

구조 The Placement I: initial, M: medial, and F: final



Other vowels are used in the 1st placement. * Marked vowels are used in the 2nd placement. ♥ Marked vowels are used in the 3nd placement.

초성 The Initial

ㄱ g ㄲ kk ㅋ k
ㄴ n
ㄷ d ㄸ tt ㅌ t
ㄹ r,l
ㅁ m
ㅂ b ㅃ pp ㅍ p
ㅅ s ㅆ ss
ㅇ No sound
ㅈ j ㅉ jj ㅊ ch
ㅎ h

중성 The Medial

ㅏ a ㅑ ya ♥ㅘ wa
ㅓ eo ㅕ yeo ♥ㅝ wo
* ㅗ o * ㅛ yo
* ㅜ u * ㅠ yu
* ㅡ eu
ㅣ i
ㅐ ae ㅒ yae ♥ㅙ wae
ㅔ e ㅖ ye ♥ㅞ we
♥ㅚ oe
♥ㅟ wi
♥ㅢ ui

Korean is even more regular than Finnish!

The symbols for similar phonemes are represented as similar to one another.

What orthographic feature corresponds roughly to velar place of articulation, for example?

종성 The Final

ㄱ ㄲ ㅋ ㄳ k
ㄴ ㄵ★ㄶ n
ㄷ ㅌ t
ㄹ★ㄻ ㄽ ㄾ★ㅀ l ★ㄺ k ㄻ m ㄿ p
ㅁ m
ㅂ ㅍ ㅄ p
ㅅ ㅆ t
ㅇ ng
ㅈ ㅉ t
ㅎ t

ㄴ(n)/ㅀ(l) +ㄱㄷㅈ -> ㄵ()/ㄹ(l)+ㅋㄷㅊ
ㄴ(n)/ㅀ(l) +ㅅ -> ㄴ(n)/ㄹ(l)+ㅆ

ㄺ(k) + ㄱ -> ㄹ(l)

ㄼ(l) of 밟 +ㅂ aeum -> ㅂ(p)
ㄼ(l) of 넓 -> ㅂ(p)

Decoding in action

Read these “pseudowords” aloud:

GLORP

SPLING

JONK

FEESH

QUISP

Decoding in action

Table 3
Mean Response Times (RTs) and Error Rates (ERs) in Experiment 2 as a Function of Stimulus Type and Blocking Condition

	Condition					
	Pure		Mixed		Effect	
Stimulus type	RT	ER	RT	ER	RT	ER
HF words	442	1.4	469	1.4	+27	0.0
LF words	482	2.2	496	2.9	+14	+0.7
Nonwords with HF words	541	6.2	520	9.8	−21	+3.6
Nonwords with LF words	552	6.6	532	6.9	−20	+0.3

Note. The mixed conditions were created by mixing words from a particular frequency class with nonwords. Thus, words from different frequency classes were not mixed with one another. HF = high frequency; LF = low frequency.

Pseudowords (here called “nonwords”) can be read nearly as quickly and accurately as real words.

(In this example, pseudoword naming times are between 5 and 12% slower than reading times for low-frequency words.)

This suggests that we “do” decoding fairly naturally, automatically and implicitly.

The questions we’ll consider when we discuss computational models of word reading have to do with *how decoding should be characterized.*

Specifically, we’ll discuss whether this behavior is best captured *by categorical rules* or *probabilistic constraints*.

Lupker, Brown, & Colombo (1997)

Decoding with ambiguity

Read these pseudowords aloud:

VONTH

PHANGE

STREAD

SOUGH

Decoding with ambiguity

VONTH

PHANGE

STREAD

SOUGH

Generalization is a good way to test what kinds of regularities people have learned about a system.

Remember, e.g., The Wug Test…(or if you don't remember, we'll get to it next week)

It's hard to imagine a rule set that covers our intuition that both /strɛd/ and /strid/ are acceptable for STREAD.

Decoding effects for real words: the "regularity effect"

Memory & Cognition
1980, Vol. 8 (5), 424-432

Lexical access and the spelling-to-sound regularity effect

DAVID W. BAUER
University of California, San Diego, La Jolla, California 92037

and

KEITH E. STANOVICH
Oakland University, Rochester, Michigan 48063

Table 2
Mean Reaction Times (in Milliseconds) for Experiment 4

	Normal		Degraded	
	Mean	ER	Mean	ER
Regular	661	3.1	712	5.4
Exception	699	5.4	753	7.6

Note–ER = error rate.

We'll examine arguments about the nature of decoding processes next time.

It's uncontroversial that some form of decoding is part of normal reading, and that words that are "irregular" (or "inconsistent") are harder to recognize than words that are regular.

This is straightforward for items like PINT, ACHE, CHOIR, etc.

It's less clear-cut for "neighborhoods" of words like ROLL/POLL/TOLL/DOLL or COOK/TOOK/BOOK/SPOOK.

So, we have a lot of evidence for the use of *sublexical* information in reading, especially in the translation from spelling to sound

People read nonwords quickly and accurately.

Words that follow typical spelling-to-sound patterns are easier to read than ones that don't.

What about *lexical* processes?

Well, there's this again.

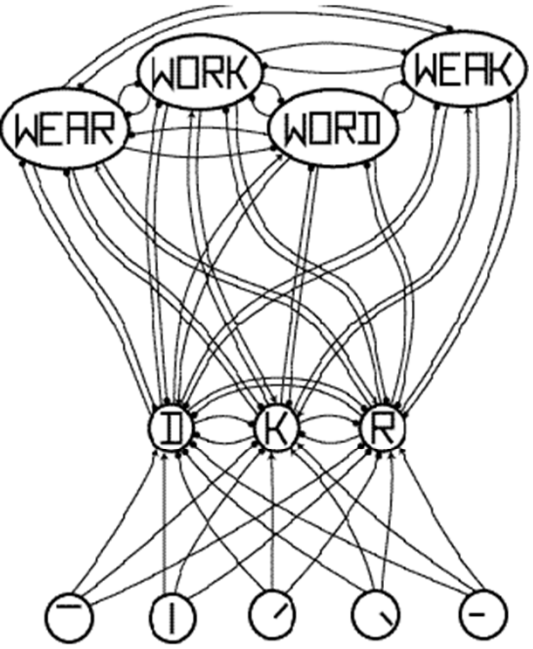


Figure 1. Interactive Activation Network Model (after McClelland and Rumelhart, 1981).



Figure 5. A hypothetical set of features that might be extracted on a trial in an experiment on word perception.

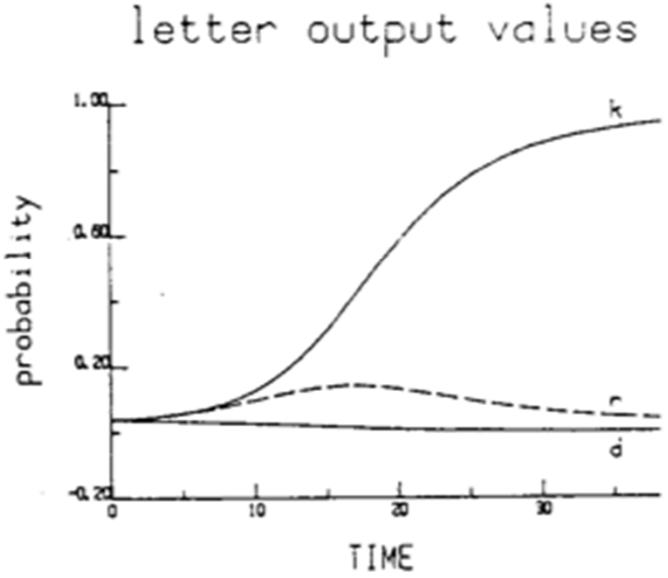


Figure 7. Output values for the letters *r*, *k*, and *d* after presentation of the display shown in Figure 5.

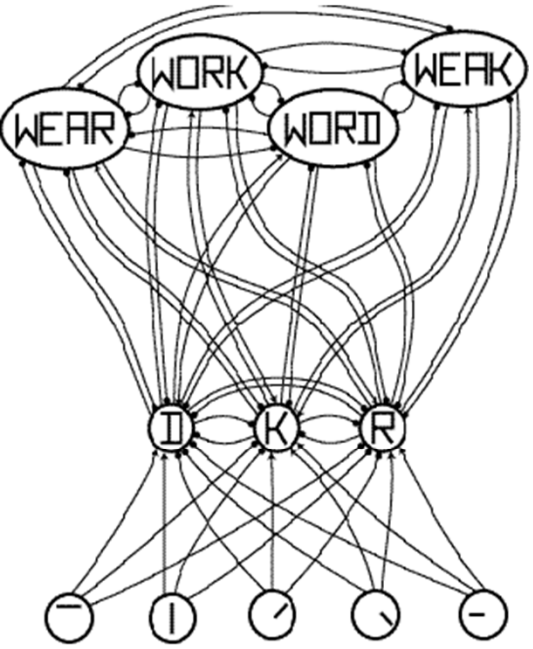


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The Interactive Activation Model:

- feedback from activation for known words increases the activation for letters that occur in those words



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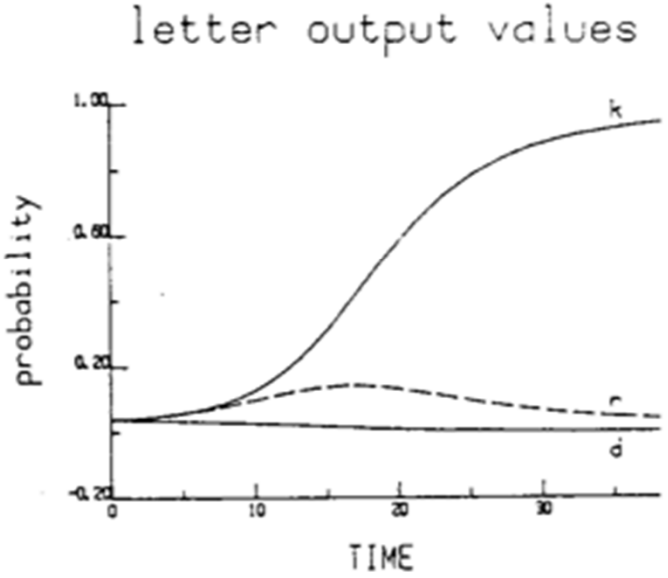


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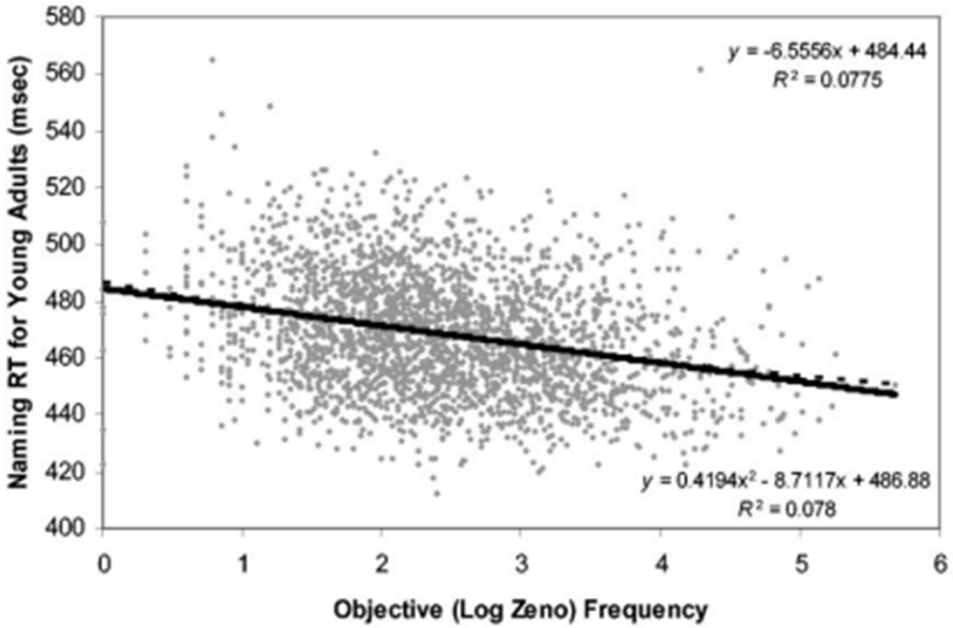
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Note. The mixed conditions were created by mixing words from a particular frequency class with nonwords. Thus, words from different frequency classes were not mixed with one another. HF = high frequency; LF = low frequency.

Also note that we've already seen **two** lexical effects:

- the overall effect of *lexicality*

- the effect of *word frequency.*



Just as with picture naming, the number of times you've encountered a word in the past has an impact on how rapidly you can name it.

This effect is robust over a very very large number of observations

(Balota et al. 2004, data from the English Lexicon Project)

Table 5

Standardized Reaction Time Regression Coefficients From Steps 1 and 2 of the Item-Level Regression Analyses for Young and Older Adult Lexical Decision Task (LDT) and Naming Performance

	Young		Older	
Predictor variable	LDT	Naming	LDT	Naming
Step 1				
Affricative	−0.18	−0.39***	−0.24†	−0.27*
Alveolar	0.44	1.18***	0.43	1.06***
Bilabial	0.39	1.03***	0.38	0.90**
Dental	0.13	0.37***	0.10	0.31**
Fricative	−0.26	−0.76**	−0.29	−0.41
Glottal	0.18	0.27*	0.10	0.18
Labiodental	0.25	0.63***	0.23	0.49**
Liquid	−0.32	−0.89***	−0.44†	−0.64**
Nasal	−0.20	−0.72***	−0.21	−0.44**
Palatal	0.35	0.57**	0.35	0.56**
Stop	−0.36	−1.12***	−0.52	−0.82**
Velar	0.39	1.03***	0.45†	0.92***
Voiced	0.10***	−0.12***	0.13***	−0.01
R^2	.01	.35	.04	.22
Step 2				
Length	−0.00	0.16***	0.07**	0.18***
Objective frequency	−0.32***	−0.13***	−0.37***	−0.20***
Subjective frequency	−0.38***	−0.13***	−0.21***	−0.13***
Neighborhood size	0.02	−0.10***	0.08***	−0.05*
Feedforward onset consistency	−0.07***	−0.03†	−0.04†	−0.03†
Feedback onset consistency	0.00	−0.08***	−0.02	−0.10***
Feedforward rime consistency	−0.05**	−0.08***	−0.05**	−0.08***
Feedback rime consistency	−0.02	−0.08***	−0.03†	−0.07***
R^2	.42	.49	.34	.39

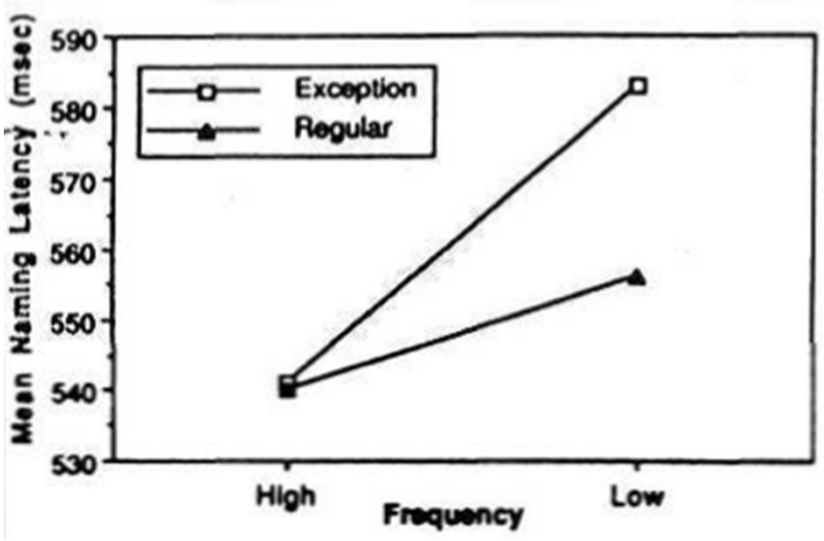
† $p < .10$. * $p < .05$. ** $p < .01$. *** $p < .001$.

Remember we worried about whether frequency effects might be confounds of length and other variables.

This appears not to be the case in a multiple regression analysis of about 3000 words, read by 30 young and older adults.

Behold: The Frequency by Regularity interaction.

The frequency effect is much larger for exception words than regular words.



This is a landmark finding in psycholinguistics.

We'll see how it has impacted a number of other areas (most immediately inflectional morphology).

What does it mean?

One interpretation

There are two separate processes in word recognition:

1. A sublexical process (decoding), and

2. A lexical process (knowledge of a particular word)

For high frequency (HF) words, the lexical process is so fast and efficient that there is little influence of the sublexical process.

The sublexical process only shows up when we read nonwords, or low frequency (LF) words.

Neuropsychological evidence for these "dual routes"

Patients with *acquired phonological dyslexia:*

- present with reading difficulty as the result of brain injury (thus "acquired" as opposed to "developmental" dyslexia)
- have difficulty *specifically with pseudoword naming.*

TABLE 3

Number and Percentage of Correct Responses Provided by M.O. and I.B. in Reading Aloud Words With Different Characteristics

Characteristic	M.O.	I.B.
Concreteness		
Concrete	19/20 (95%)	19/20 (95%)
Abstract	18/20 (90%)	17/20 (85%)
Grammatical class		
Nouns	25/26 (96%)	26/26 (100%)
Verbs	24/26 (92%)	25/26 (96%)
Adjectives	25/26 (96%)	25/26 (96%)
Functors	25/26 (96%)	25/26 (96%)
Frequency		
High	25/25 (100%)	25/25 (100%)
Low	25/25 (100%)	25/25 (100%)

TABLE 4

Numbers of Plausible Responses Provided by M.O. and I.B. in Reading Nonwords Aloud

Word List	M.O.	I.B.
Woodcock & Johnson (1989): Word Attack	12/28 (43%)	12/28 (43%)
Friedman, Ferguson, Robinson, & Sunderland (1992)	27/69 (39%)	18/69 (26%)
Glushko (1979)	66/138 (48%)	58/138 (42%)
Kay & Patterson (1985)	47/80 (59%)	22/80 (27%)
Berndt, Haendiges, Mitchum, & Wayland (1996)[a]		
High grapheme-to-phoneme correspondence	27/32 (84%)	21/32 (66%)
Low grapheme-to-phoneme correspondence	4/20 (20%)	3/20 (15%)

Cacciopo van Vliet et al. (2003)

Neuropsychological evidence for these "dual routes"

Patients with "acquired surface dyslexia:"

- also present with reading difficulties after brain injury

- difficulties are specifically with *irregular words.*

- mispronunciations tend to be "regularizations."

	Correct Performance					
Patient	HFR	LFR	HFE	LFE	%Reg's	NW
MP	95	98	93	73	90	95.5
KT	100	89	47	26	85	100

DEAF ⇒ "deef"
SAID ⇒ "sayed"
BROAD ⇒ "brode"
SHOE ⇒ "show"
ONE ⇒ "own"

FLOOD ⇒ "flude"
GONE ⇒ "goan"
STEAK ⇒ "steek"
SEW ⇒ "sue"
SOOT ⇒ "suit"

Neuropsychological evidence for these "dual routes"

Patients with *acquired phonological dyslexia:*

- present with reading difficulty as the result of brain injury (thus "acquired" as opposed to "developmental" dyslexia)

- have difficulty *specifically with pseudoword naming.*

- *relatively spared exception word reading*

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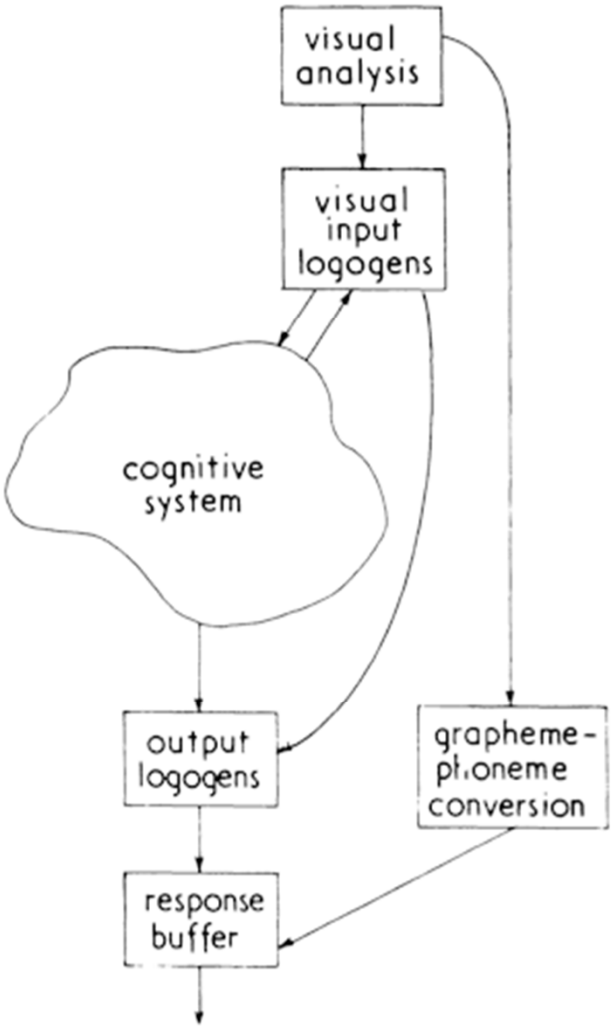
- mispronunciations tend to be "regularizations."

- *relatively spared pseudoword reading.*

This is a classic *double dissociation.*

Why are double dissociations so powerful?

Why are double dissociations so powerful?



They provide evidence that processes underlying pseudoword and irregular word reading (for example) are *separable.*

In this model, we can imagine:

- phonological dyslexia results from *specific* damage to the grapheme-phoneme conversion module

- surface dyslexia results from damage to input or output "logogens", or the connections between them

Figure 1. Part of Morton's logogen model dealing with recognition, comprehension and oral reading of written words and letter strings (from Morton & Patterson, 1980*a*, Fig. 4.26, p. 95).

Early inklings that it is not so simple…

Whatever description eventually accounts most satisfactorily for the evidence on synthesis of phonology, surface dyslexia may well be an important source of that evidence. In fact, I have dwelt rather more on this topic than on most of the theoretical issues because I find it an intriguing example of the contribution of neuropsychological observations. As described by too many writers to list, a fashionable view of oral reading in recent years has been the 'dual-route' position. Words could be recognized from a visual/graphemic code and pronounced from the stored phonological specification for the whole word; or they could be handled by non-lexical grapheme–phoneme translation. Reading errors in surface dyslexia were considered to reflect the operation of the second route, and should therefore show no 'lexical' influences. Marcel's (1980) careful scrutiny of the error data however reveals them to be subject to several such influences: in particular, as already noted, neologisms do not occur as often as a purely non-lexical process would predict. (The reading error *incense* → 'increase', for instance, can be considered an example of difficulty with the ambiguous consonant *c*; but if that were the only factor, surely the patient should have produced the neologism 'inkense'.) This poses a classic dilemma in our search for explanations: having discovered lexical effects in what were supposed to be the non-lexical reading errors of surface dyslexia, do we change our view of surface dyslexia (and conclude that it reflects a more complicated combination of the two routes) or change our view of non-lexical reading (and conclude that it represents a more complicated route)? Of particular note is the fact that theorists modelling normal reading processes have been asking the same question, for example with regard to pronunciation of non-words by normal subjects (Glushko, 1979; Henderson, in press). It is probably the confluence of normal and pathological evidence which holds the best promise for advances in understanding.

Patterson (1981, British Journal of Psychology)